NAME : UTSAV KUMAR KUSHWAHA

ROLL NO : 63

SUBJECT : COMPUTER VISION

What is Optical Flow?

Optical Flow is the paflern of apparent motion of objects between two consecutive image frames caused by object movement or camera movement.

Instead of detecting an object directly, optical flow estimates how pixels move from one frame to another.

5 There are two major types:

Sparse Optical Flow: Tracks selected feature points. Dense Optical Flow: Computes motion for every pixel. Flow Diagram Algorithm 1: Lucas-Kanade Sparse Optical Flow Principle Lucas-Kanade tracks only important feature points such as corners. It assumes: Motion is small. Neighboring pixels move together. Brightness remains constant.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
from google.colab import files

uploaded = files.upload()
video_path = list(uploaded.keys())[0]
```

Choose Files Snapchat-1…679931.mp4

Snapchat-1171679931.mp4(video/mp4) - 9408472 bytes, last modified: 1/20/2025 - 100% done

```
Saving Snapchat-1171679931.mp4 to Snapchat-1171679931.mp4
```

```
import cv2
import numpy as np

cap = cv2.VideoCapture(video_path)

feature_params = dict(maxCorners=100,
                      qualityLevel=0.3,
                      minDistance=7,
                      blockSize=7)

lk_params = dict(winSize=(15,15),
                 maxLevel=2,
                 criteria=(cv2.TERM_CRITERIA_EPS |
                           cv2.TERM_CRITERIA_COUNT,
                           10,
                           0.03))

ret, old_frame = cap.read()
old_gray = cv2.cvtColor(old_frame, cv2.COLOR_BGR2GRAY)

p0 = cv2.goodFeaturesToTrack(old_gray, mask=None, **feature_params)

mask = np.zeros_like(old_frame)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame_gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    p1, st, err = cv2.calcOpticalFlowPyrLK(
        old_gray, frame_gray, p0, None, **lk_params)

    if p1 is not None:
        good_new = p1[st==1]
        good_old = p0[st==1]

        for new, old in zip(good_new, good_old):
            a,b = new.ravel()
```

```
                c,d = old.ravel()

                mask = cv2.line(mask,(int(a),int(b)),
                                (int(c),int(d)),(0,255,0),2)

                frame = cv2.circle(frame,(int(a),int(b)),
                                   4,(0,0,255),-1)

            img = cv2.add(frame,mask)

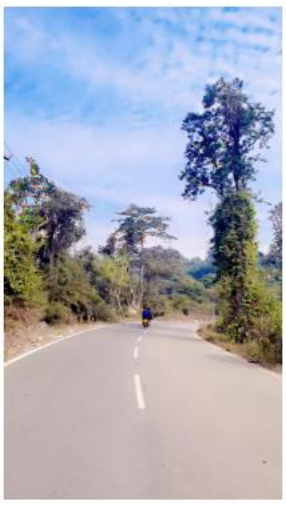
            plt.imshow(cv2.cvtColor(img,cv2.COLOR_BGR2RGB))
            plt.axis("off")
            plt.show()

            old_gray = frame_gray.copy()
            p0 = good_new.reshape(-1,1,2)

    cap.release()
```



































































```
Start coding or ge_nerate with AI.
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

cap = cv2.VideoCapture(video_path)

ret, frame1 = cap.read()
prvs = cv2.cvtColor(frame1, cv2.COLOR_BGR2GRAY)

hsv = np.zeros_like(frame1)
hsv[...,1] = 255

while True:

    ret, frame2 = cap.read()

    if not ret:
        break

    next_frame = cv2.cvtColor(frame2, cv2.COLOR_BGR2GRAY)

    flow = cv2.calcOpticalFlowFarneback(
        prvs,
        next_frame,
        None,
        0.5,
        3,
        15,
        3,
        5,
        1.2,
        0)

    mag, ang = cv2.cartToPolar(flow[...,0], flow[...,1])

    hsv[...,0] = ang*180/np.pi/2
    hsv[...,2] = cv2.normalize(
        mag,
        None,
        0,
        255,
        cv2.NORM_MINMAX)

    rgb = cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB)

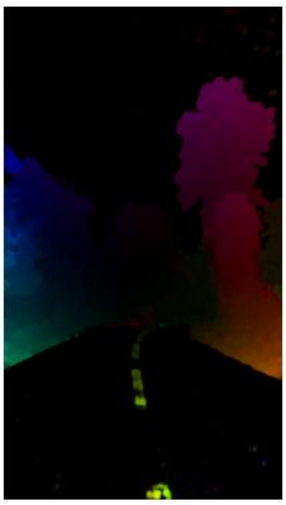
    plt.imshow(rgb)
    plt.axis("off")
    plt.show()

    prvs = next_frame

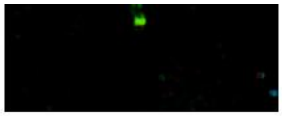
cap.release()
```

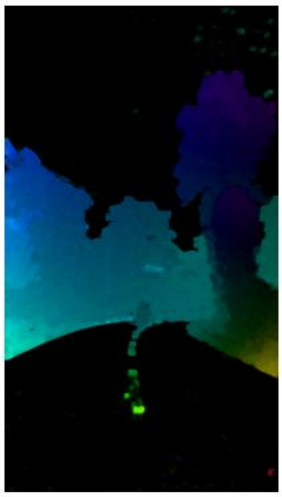




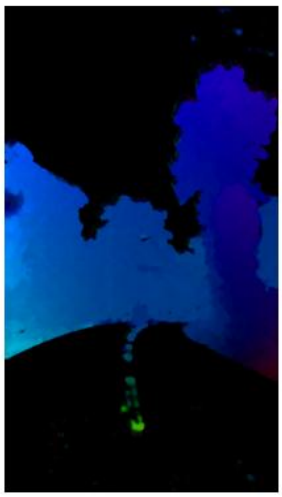


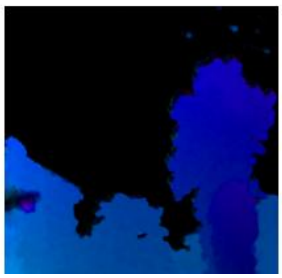


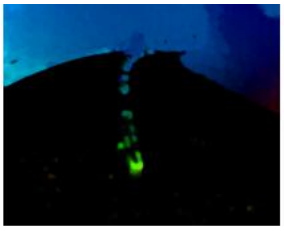


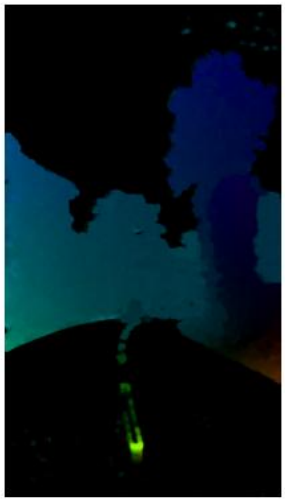


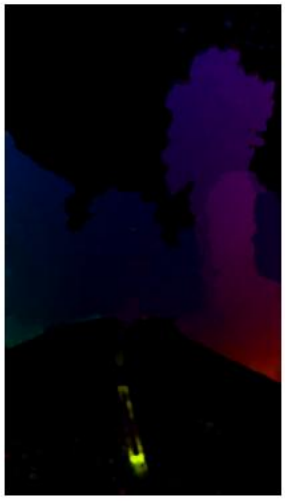


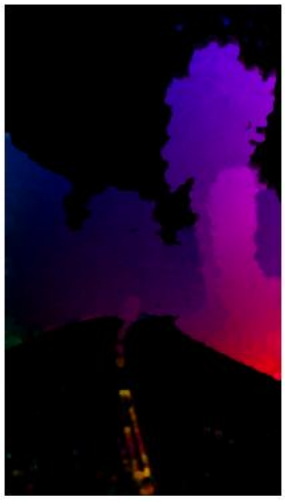














Viva Questions with Answers

Q1. What is Optical Flow?

Optical Flow is a technique that estimates the movement of pixels between consecutive video frames. It is used for motion detection, object tracking, and scene understanding.

Q2. Explain the Lucas-Kanade algorithm.

Lucas-Kanade assumes that pixel brightness remains constant and nearby pixels move similarly. It detects feature points and tracks them across frames.

Q3. What is Dense Optical Flow?

Dense Optical Flow estimates motion for every pixel in an image, unlike Sparse Optical Flow which tracks only selected feature points.

Q4. Compare Lucas-Kanade and Farneback.

Lucas-Kanade	Farneback
Sparse	Dense
Fast	More detailed
Tracks feature points	Tracks all pixels

Q5. What assumptions are made?

- Brightness remains constant.
- Motion between frames is small.
- Neighboring pixels move similarly.

Q6. Factors affecting accuracy

- Lighting changes
- Motion blur
- Fast movement
- Camera shake
- Noise

Q7. Five applications

1. Autonomous driving
2. Video surveillance
3. Gesture recognition
4. Sports analytics
5. Robotics

Q8. Why use grayscale images?

Grayscale images reduce computation while preserving intensity information required for motion estimation.

Q9. Limitations

- Sensitive to illumination changes.
- Difficulty with occlusions.
- Reduced accuracy for large motion.
- Noise affects performance.
- Camera movement introduces false motion.

Q10. Contribution to autonomous driving and surveillance

Optical Flow helps detect moving vehicles, pedestrians, and objects, enabling tracking, collision avoidance, traffic monitoring, and action recognition in intelligent vision systems.